

Research Guide

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Chapter 1 Introduction

Doing research is like baking: it requires structure (you need to follow a series of steps), but it also requires passion and some magic. It requires [Deep Work](#), and a clear commitment to produce something that helps us understand the world a little bit better.

In this Guide, I will introduce some of the elements that will help you produce a great research project. Just as with a baking book, I will introduce some of the elements and stages you need to follow in producing your research paper. But it is important that you also feel free to include your own magic. Use the elements that will help you keep motivated in the process and make it feel your own.

In [Section 1](#), I start by providing a general overlook of what makes a research project, a *great* one. Then, [Section 2](#) looks into the most basic issue: **How to find a research question?**. Even the best research project will be useless if the research question does not attract the attention of an audience. The Guide introduces what are some of the main characteristics of a *great* research question, and how to get inspired. I include some handouts with some of the stages that you need to go through in order to find the best research question for you.

Then, in [Section 3](#), I will look into some of the most important aspects that you need to keep in mind when structuring your argument. [Section 4](#) introduces some of the elements that every empirical paper (in economics and political science) that follow quantitative research methods needs to include. [Section 5](#) mentions some of the issues to keep in

mind when writing an empirical model and working with data. [Section 6](#) focuses on how to structure the presentation of your research project and provides some tips to keep in mind when presenting.

This volume, of course, is still a work in progress. It compiles information that I have found useful from multiple scholars. I include the specific citations and links in the specific sections, so you can read more about it. It also includes some things that I learned while working on my own research (most of it, I have learned the hard way). But, as I gain more experience and find more information, I will expand this Guide¹.

But, above all, my main message is that research is fun. It allows you to explore and learn more about a topic and contribute to our understanding of how the world works. It also provides you with a set of skills that will be useful in your future careers and hopefully, in your lives. You will be able to tell a story in a clear and effective way, you will learn how to analyze data, and how to become a better writer and communicator. I hope that you learn that economics provides you with a set of tools and knowledge that can be combined with information and knowledge in other fields, and that you belong in this process of improving our knowledge in the discipline.

Now, let's do some research!

Chapter 2 Where to Start?

Working on a research project can be frightening. The papers that we read at school are sophisticated, have sound theories that support the empirics, the data is rich, and the model seems solid. When we read papers that attract our attention, it seems as if all the good questions have been asked. Finding a research question seems like an overwhelming task.

In general, the quality of a research project is a function of the quality reached at different stages of the process. In my view, the following are the most crucial stages of a research project.

1. Formulating a good research question
2. Finding appropriate data for the project
3. Using a sound theoretical model
4. Designing a clear empirical approach
5. Writing with clarity

Paying close attention to the tasks and details that each stage requires will allow us to reach high quality in the final product. This may seem daunting at first, but having a clear structure of the tasks we need to complete will make the process much more enjoyable.

The most important element, in my opinion, is to realize what we are trying to accomplish with a research project. You may be interested in passing a class, and therefore you need to complete a research project. But, beyond that, the main objective of any research project is to understand one small area in our world a little bit better. That must serve as a motivator.

In social sciences we use the scientific method to attain this goal:

1. We start with a general theory: *x causes y*
2. We develop a hypothesis around this theory: *An increase in x will lead to an increase in y*
3. We then collect data that will help us test this hypothesis
4. We analyze the data and the results.
5. We *accept* or *do not accept* our hypothesis.
6. If pertinent, we expand the theory to different issues.

We will now analyze how to start thinking about a research question that can help us understand the world a little bit more and that follows the scientific method.

Chapter 3 The Research Question

A good paper starts with a good research question. But how to identify a **great** research question? In this Module, I will introduce some of the characteristics that make a question a good research question. This section draws on Nick Huntington-Klein's [The Effect](#).

3.1 Elements to Consider

A great research question needs to fulfill a set of characteristics:

1. *It is based on theory*: Before even having data collected on this, you need to have a good story behind the potential outcomes you obtain. There must be something interesting to say about what you did.
2. *It is feasible*: This characteristic is relevant for different reasons. First, you need to know that the data you need to answer the question is available, or that you can collect it. Second, even if it is possible to obtain the data, you need to be certain that the data is actually going to help you answer the question you have. Third, time is limited, so you also need to consider if you can answer the question within the time frame you considered. Finally, you also need to consider if there is a research design that will allow you to answer your research question. This particularly happens with questions that are more convoluted.
3. *It is relevant*: In the past, I was advised to focus on questions that affect at least a 3% of the world population. Although I believe that number is arbitrary and not necessarily useful, I think that the main point remains. Choose questions that are relevant for at least one community (Los Angeles, San Bernardino, a set of small firms in one region, etc). Of course, projects that address issues that affect a larger share of the population are sometimes easier to address, as data tends to be more available, but you also need to consider novelty, which is my next point.
4. *It has not been answered before*: This point requires some clarification. Researchers build upon each other's work. We do not reinvent a topic each time. Knowledge progresses at a marginal rate. To do that, research projects do one of three things:

- Contradict a current theory.
- Extend a theory to a new sample to reach external validity (or at least suggest internal validity in a new context).
- Use a different method to test a theory. This is, you do not need to reinvent the wheel with your research project, but need to be original in how you answer the question.

3.2 Identifying Questions

The first step to find your research topic is to think deeply about the issues you care about. These issues, of course, need to satisfy the previous characteristics I introduced. For instance, you may care about your dog Fluffy, and you may want to know what makes him so adorable, but that is not (a) a relevant question for a large share of the population; (b) answerable (you do not have a large sample).

One way to identify potential research topics is by thinking the topics you analyzed in your courses and that can be addressed in an economic way. Think about the courses you liked the most, or a particular topic in a class. This may actually help since probably you already identified some literature on the topic. But remember, the question needs to be analyzed with economic lenses, and needs to fulfill the characteristics that I mentioned previously.

Another way to identify a question is to link it to your future (aspired) career. This, of course, requires that you have a clear idea on what you would like to do in the future, and also needs for you to be original in the way you tackle a subject.

This leads me to my next point. It is hard to identify questions if you do not know what is available, what has been done, and how. This is why you need to read and be curious. You need a critical mind. Just as artists do, obtain inspiration from different, and sometimes unusual, sources. When I ask you to read, this is not limited to academic literature. In fact, a large proportion of what I read is non-academic. Of course we all have a bias on what we read and see based on our interests, but try to broaden your spectrum, you may be surprised about how many interesting things exist out there.

3.3 Determinant Questions

Sometimes, we may be interested in a topic and want to learn and respond everything about it. We get so excited about what are the causes behind a certain issue, that we focus on everything at once.

However, as [Marc Bellemare](#) says it, good social science follows the scientific method. This is, you need to have a hypothesis “(x causes Y)”, then test it, and finally analyze it.

Determinant papers do not follow this method. Instead, you use a bunch of variables (you have many x’s), but you lack a research design, and therefore just look at correlations, without a clear argument behind it. This type of papers may be useful in some cases (you can read another blog entry from [Marc Bellemare](#) on this, or look at the [Simpson Paradox](#) on Wikipedia).

You may decide to later on follow a given research agenda, but for one research paper (and particularly, for your senior thesis), keep your question narrow and specific. Briefly, *avoid determinant questions!*

This also includes avoid giving a clear interpretation to control variables and use them as secondary research questions. Instead, you may want to use a subsample of your data, or disentangle some of the effects behind your main variable of interest.

3.4 Some Examples ²

Not-so-Good Research Question	Good Research Question
What are the determinants of migration?	How does climate change affect migration?
What causes criminal behavior?	How does access to public health affect criminal behavior?
How does business competition work?	How do incumbent companies respond when a new company enters the market?
Is education good?	What are the long-run effects of attending prep school?
What causes economic growth?	What is the effect of trade liberalization in promoting economic growth?

3.5 Handouts

- [Research Question](#)
- [Calendar-Spring 2022](#)

Chapter 4 Research Writing

4.1 Structure

In this section, I focus on the main stages of the research writing process. Most of these concepts have been beautifully explained by [Varanya Chaubey \(2018\)](#). We will be focusing on the book, but in this section, I compile some of the most interesting ideas and link them to other important aspects to consider when structuring an argument. Some of this material is structured with more detail on Laura Belcher's book [Writing your Journal Article in Twelve Weeks](#).

4.2 The Three Layer Method

Once we have found our research question and we obtained and processed the data we need to conduct our analysis, we need to write our results.

This method asks us to work from the general ideas to the details, using a *descending structure*, or a **Three layer method**.

This method is a 3-step process in which we start working by laying a foundation of the main project and build upon it. The concept is simple: we need to understand what we are doing, why and how before even immersing in the writing process. Otherwise, we will lose sight of the main objective. The process is straightforward and quite intuitive. I introduce the three stages of the process here and explain each of them below.

- **Step 1:** What are you saying?: This is the main argument that you are making. It is important to figure out if you actually have an argument. But I'll come back to this point.
- **Step 2:** Express with an outline. You need to include additional information surrounding your argument, so the readers can answer follow-up questions and have

additional details linked to your research question.

- **Step 3:** Develop your ideas in a draft. Once you have identified your main argument and have an outline, you need to structure the paragraphs in each section.

4.2.1 The Argument

Belcher (2019) defines an argument as: *“your article’s most important idea stated in one or two sentences early and clearly in your article [...], emerging from a theory and supported with evidence to convince the reader of its validity.”*

This may sound trivial, but it is harder than it seems. Many times, we believe we already have an argument, but we really do not. Instead, we have sentences that are tautological or we are simply rephrasing a fact that is accepted by everyone. Therefore, Belcher proposes a set of tests to ensure that you actually have an argument (I am adapting the list for the purposes of this Guide):

1. **Agree/disagree:** Do we need evidence to agree or disagree with a particular statement? For instance, we do not need further evidence to the statement ‘The Earth is round’. But we may need evidence on the statement “Prep school is fundamental to children’s cognitive development.”
2. **Dispute test:** When a given statement can be the source of disagreement, then it seems that you may indeed have an argument. For instance, “Poorer people are less supportive of redistribution” (AEP, 2021)
3. **Puzzle answer test:** If your statement is providing a response to a question that people have about the world or their environment, you may have an argument.

Another important element is to differentiate your argument from your topic. The topic is the major issue you are interested in, whereas your argument explains the main finding (or initially, the hypothesis) of your paper.

Following the research question, an argument needs to be puzzling. It needs to provide relevant information that help us understand the world a little bit more. This is why your argument (as well as your research question) needs to go beyond the basic facts. It needs to provide enough detail as to make it interesting for a larger audience. This also entails

that you need to provide more information than naming the main variables in your analysis (x causes Y). You need to specify the conditions and context that make this statement to hold.

Some other elements to consider when structuring your argument is to avoid including *normative* statements and speculations, More specifically, for quantitative papers:

Avoid including causal claims when the evidence does not allow you to do that.

Causal analysis is key in our field, but correlations are important as well and they provide a value to understand our context a little bit more.

4.2.1.1 Finding your RAP

1. **R:** Have different versions of your research question to see what is the clearest way to introduce it to your readers.
2. **P:** This represents how you position the paper in the literature. This is constructed based on your literature review and the theory behind your question.

These three elements are interconnected. You need to find the best way to bring them all together and work with them to convey your argument.

4.2.2 Express your Ideas using an Outline

An empirical, quantitative, paper in economics (and political science) usually contains the following sections:

1. Abstract
2. Introduction
3. Context (Literature Review) 4a. Theoretical papers contain mathematical models (we will not use those) 4b. Empirical Strategy
4. Results
5. Robustness checks and potential mechanisms (we will not focus on those)
6. Final discussion (Conclusion)

We will talk more about each of these sections, but here, the main point to consider is that you need to create an outline that conveys the most important points of each section.

This is, after you have a clear argument, now you need to provide an answer to different questions that the readers may have. This is done by creating the headings and subheadings of each section. For instance, in a paper on mining in the Democratic Republic of the Congo (DRC), readers may be interested in learning why is mining important in the country and what types of mining take place in the country. This means that I need a general section on the context of mining in the DRC and then include subheadings explaining the different types of mining that I analyze.

You will do that for each section. In your outline, include the headings and subheadings, and a short paragraph indicating the main message of the section. This will then be enriched by secondary paragraphs.

Having this structure will allow you to include those sections that add value to your final paper and remove any additional information that is not key to support your main argument.

4.2.2.1 Drafting

Once you have your headings and subheadings, as well as the most important takeaways, it is time for you to start populating your paper. In the next section, I mention some of the elements that you need to include in the research paper. Here again, it is important that you plan the information that you will include and that each paragraph has a purpose, answering a question that is relevant to further your argument. Go for the general to the particular details.

The main thing to consider is that readers have very limited time and span of attention. You need to convey the main message at the beginning of the paper. Then, for each section, the main idea needs to be included in the first paragraph(s). Develop just one idea per paragraph and ensure that the main message is contained at the beginning.

Writing is an iterative process and you probably will spend more time rewriting a section than what you spent writing it for the first time. Don't despair! We all go through the same process and you will get there. Just ensure that you structure and organize your process.